192. What type of vibrations are produced in a sitar wire?
(a) Longitudinal vibration
(b) Transverse vibration
(c) Both (a) and (b)
(d) None of the above

193. Consider the following statements
1. Waves on springs or sound waves in air are longitudinal waves.
2. Waves on surface of water is mechanical waves.
3. Mechanical waves are used in communication, telephony
4. Mechanical waves can propagate in solid, liquid and gas.

Which of the following statements is/are correct.
(a) 1, 2 and 4 (b) 1 and 2
(c) 2 and 4 (d) Only 3

194. The most familiar form of radiant energy in sunlight that causes tanning and sunburning of human skin, is called
(a) ultraviolet radiation
(b) visible radiation
(c) infrared radiation
(d) microwaves radiation

195. Microwaves are
1. used in photography.
2. absorbed by water molecules and living tissue internal heating will damage or kill cells.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) Neither 1 nor 2

196. Match the following columns.

Column I	Column II
A. Infrared rays	1. Medical tracers
B. γ-rays	2. TV control
C. Radio waves	3. Making ions
D. Ultraviolet rays	4. Communication

Codes

	A	B	C	D		A	B	C	D
(a)	3	2	1	4	(b)	2	1	4	3
(c)	3	1	2	4	(d)	3	4	2	1

197. Sound waves travel with a speed of about 330 m/s. The wavelength of sound whose frequency is 550 Hz is
(a) 0.8 m (b) 8.4 m (c) 0.6 m (d) 0.2 m

198. Which one of the following statements is correct?
(a) Sound waves in air are longitudinal while light waves are transverse
(b) Sound waves in air are transverse while light waves are longitudinal
(c) Both sound and light waves in air are transverse
(d) Both sound and light waves in air are longitudinal

199. Microphone is a device in which
(a) amplification is not required at all
(b) sound waves are directly transmitted
(c) sound waves are converted into electrical energy and then reconverted into sound after transmission
(d) electrical energy is converted into sound waves

200. Sound travels faster when the medium is of
(a) low density (b) medium density
(c) high density (d) vacuum

201. A source of sound has a noise level of 20 dB. If the intensity of sound produced by the source increases by 100 times, the noise level becomes
(a) 30 dB (b) 40 dB
(c) 400 dB (d) 20000 dB

202. Which among the following statement is correct regarding the audible sound waves?
1. Its frequency is greater than 20000 Hz.
2. It is used in measuring the depth of sea.
3. The liquid from which the audible sound waves pass gets hot.

The correct statement(s) is/are
(a) 1 and 3 (b) 1 and 2
(c) Only 1
(d) None of the above

203. Match the following columns.

Column I (Waves)	Column II (Sources)
A. Ultrasonic waves	1. Ocean waves
B. Sound waves	2. Energetic disturbance
C. Infrasonic waves	3. Generated by vibrating bodies
D. Shock waves	4. Wave produce by bat

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	3	2	1	4
(c)	4	3	2	1	(d)	1	2	3	4

204. Which of the following statement is incorrect?
(a) Sound travels in straight line
(b) Sound is a form of energy
(c) Sound travels in the form of waves
(d) Sound travels faster in vacuum than in air

205. Sound travels in rocks in the form of
(a) longitudinal elastic waves only
(b) transverse elastic waves only
(c) both longitudinal and transverse elastic waves
(d) non-elastic waves

206. Ultrasonic waves are those waves
(a) to which man can hear
(b) man cannot hear
(c) are of high velocity
(d) of high amplitude

207. Consider the following statements
1. The frequency of sound waves is 20 Hz to 20000 Hz.
2. Sound waves are transverse mechanical waves.
3. Sound waves travel in vacuum.
4. Sound waves have low frequency and high wavelength.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2, 3 and 4
(c) 1 and 4 (d) Only 4

208. Ultrasonic waves are used
1. to investigate the action of heart.
2. for sending signals.
3. measuring the depth of sea

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) All of these

209. Consider the following statements
1. Speed of sound basically depends upon elasticity and density of medium .
2. Speed of sound is changed by the increase or decrease of pressure
3. The speed of sound is more in humid air than in dry air.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) None of these

210. A woman's voice is shriller than a man's due to
(a) higher frequency
(b) higher amplitude
(c) lower frequency
(d) weak vocal chords

211. The phenomenon of beats can take place
(a) for longitudinal waves only
(b) for transverse waves only
(c) for both longitudinal and transverse waves
(d) for sound waves only

212. Doppler's effect is used
1. to determine the velocities of which stars and galaxies are moving toward or away from the earth
2. to increase the pressure.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

Previous Years' Questions

213. The time period of a simple pendulum having a spherical wooden bob is 2 s. If the bob is replaced by a metallic one twice as heavy, the time period will be 2012 I
(a) more than 2s (b) 2s
(c) 1s (d) less than 1s

214. Two identical piano wires have same fundamental frequency when kept under the same tension. What will happen if tension of one of the wire is slightly increased and both the wires are made to vibrate simultaneously? 2012 I
(a) Noise (b) Beats
(c) Resonance (d) Non-linear effects

215. In SONAR, we use 2012 II
(a) ultrasonic waves
(b) infrasonic waves
(c) radio waves
(d) audible sound waves

216. Motion of an oscillating liquid column in a U-tube is 2013 II
(a) periodic but not simple harmonic
(b) non-periodic
(c) simple harmonic and time period is independent of the density of the liquid
(d) simple harmonic and time period depends on the density of the liquid

217. An oscilloscope is an instrument which allows us to see waves produced by 2014 II
(a) visible light (b) X-rays
(c) sound (d) γ-rays

218. Which one of the following statements is correct?
The velocity of sound 2016 I
(a) does not depend upon the nature of media
(b) is maximum in gases and minimum in liquids
(c) is maximum in solids and minimum in liquids
(d) is maximum in solids and minimum in gases

219. Which one of the following statements is not correct? 2016 I
(a) Sound waves in gases are longitudinal in nature
(b) Sound waves having frequency below 20 Hz are known as ultrasonic waves
(c) Sound waves having higher amplitudes are louder
(d) Sound waves with high audible frequencies are sharp

220. A person rings a metallic bell near a strong concrete wall. He hears the echo after 0.3 s. If the sound moves with a speed of 240 m/s, how far is the wall from him? 2016 I
(a) 102 m (b) 11 m (c) 51 m (d) 30 m

Optics

221. The mirror used in vehicles as rear view mirror is
(a) convex mirror (b) concave mirror
(c) plane mirror
(d) plano-concave mirror

222. Viewfinders, used in automobiles to locate the position of the vehicles behind, are made of
(a) plane mirror (b) concave mirror
(c) convex mirror (d) parabolic mirror

223. Concave mirrors are used
1. as reflectors in automobiles and hand torches.
2. in the field of solar energy to focus sun's ray for heating solar furnaces.
3. as rear-view mirror in automobiles.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 3 and 1 (d) Only 1

224. A coin immersed in water pond appears to be raised, when viewed from the top. What is this due to?
(a) Total internal reflection of light
(b) Scattering of light
(c) Reflection of light
(d) Refraction of light

225. Sun appears reddish during the rising and setting time, because
(a) the atmosphere absorbs short wavelengths more than long wavelengths
(b) red light is emitted in huge amount by it
(c) the atmosphere absorbs long wavelength more than short wavelengths
(d) light of shorter wavelengths are scattered to a greater extent than the longer wavelengths by the atmosphere

226. The phenomenon of mirage is due to
(a) change in refractive index of air with change in temperature
(b) total internal reflection
(c) absorption of light by air at high temperature
(d) polarisation of light on reflection

227. Which one of the following phenomenon cannot be attributed to the refraction of light?
(a) Twinkling of stars
(b) Mirage
(c) Rainbow
(d) Red shift

228. Match the following columns.

Column I	Column II
A. Intensity of light	1. Properties of the medium
B. Colour of light	2. Refractive index of medium
C. Velocity of light	3. Amplitude of light
D. Propagation of light	4. Frequency of light

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	2	1	(b)	2	4	1	3
(c)	3	1	2	4	(d)	4	2	3	1

229. Which of the following illustrations is of refraction?
1. Twinkling of stars
2. Oval shape of sun in the morning and evening
3. Sparking of diamond
4. Mirage and looming

Which of the following statements is/are correct?
(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) Only 4

230. Efficient working of optical fibre is dependent on which one of the following?
(a) Reflection
(b) Interference
(c) Diffraction
(d) Total internal reflection

231. Condition of total internal reflection
1. Light must be propagating from denser to rarer medium.
2. The critical angle must exceeds the angle of incidence.

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) 1 and 2
(d) None of the above

232. Where should an object be placed, so that a real and inverted image of the same size is obtained using a convex lens?
(a) Between the lens and its axis
(b) At the focus
(c) At twice the focal length
(d) At infinity

233. A thin lens has a focal length of –25 cm. Then, the nature of lens is
(a) concave lens of power 4 D
(b) convex lens of power 4 D
(c) convex lens of power 25 D
(d) convex lens of power $\frac{1}{4}$ D

234. Parallel light rays entering a convex lens always converge at
(a) centre of curvature
(b) the focal plane
(c) the principal focus
(d) a point on the principal axis

235. We always get real image in a convex lens, if object is situated beyond
(a) optical centre
(b) focus
(c) centre of curvature
(d) None of the above

236. The power of a lens of focal length 80 cm is
(a) 80 D (b) 40 D (c) 12.5 D (d) 1.25 D

237. A diffraction pattern is obtained using a beam of red light. Which one among the following will be the outcome, if the red light is replaced by blue light?
(a) Bands disappear
(b) Diffraction pattern becomes broader and further apart
(c) Diffraction pattern becomes narrower and crowded together
(d) No change

238. Dispersion produced by a prism depends on
(a) its refracting angle
(b) size of the prism
(c) height of the prism
(d) one of the angle at the base of the prism

239. When dispersion of white light takes place through a prism, the maximum deviation is suffered by which one of the following colour?
(a) Red (b) Yellow
(c) Green (d) Violet

240. In the visible spectrum, the colour having the shortest wavelength is
(a) blue (b) yellow
(c) red (d) violet

241. Two beams of light, one red and the other green, fall on the same spot on a white screen. The colour on the screen will appear to be
(a) magenta (b) blue
(c) yellow (d) cyan

242. Consider the following statements
1. The colour of the green flower seen through red glass appears to be dark.
2. Red glass transmits only red light.

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

243. The focal length of convex lens is
(a) the same for all colours
(b) shorter for blue light than for red
(c) shorter for red light than for blue
(d) maximum for yellow light

244. Consider the following statements
1. The colour of an object is determined by the colour of the light reflected by it.
2. If all the colours of white light are absorbed by the object then it appears white.
3. Dangers signals are of red colour because red colour of light scatter least and therefore signals can be seen from far.

Which of the following statements is/are correct?
(a) 1 and 3
(b) Only 3
(c) Only 1
(d) 2 and 3

245. Rainbow is formed due to
1. dispersion of light
2. suffering refraction
3. total internal reflection
4. All of the above

Which of the following statements is/are correct?
(a) 1 and 2
(b) 3 and 1
(c) Only 4
(d) Only 1

246. Which one of the following sets of colour combinations is added in colour vision in TV?
(a) Yellow, green and blue
(b) White, black, red and green
(c) Red, green and blue
(d) Orange, pink and blue

247. If the light moving in a straight line bends by a small but fixed angle, it may be a case of
(a) diffraction
(b) dispersion
(c) reflection
(d) refraction and dispersion

248. If a glass rod is immersed in a liquid of the same refractive index, then it will
(a) look longer (b) disappear
(c) look bent (d) None of these

249. Why is it difficult to see through fog?
(a) Rays of light suffer total internal reflection from the fog droplets
(b) Rays of light are scattered by the fog droplets
(c) Fog droplets absorb light
(d) The refractive index of fog is extremely high

250. The colour of the sky is blue because of
(a) combination of various lights producing blue colour
(b) scattering of light by dust particles
(c) diffraction of light
(d) reflection of light by water droplets

251. Rainbow is formed due to a combination of
(a) refraction and scattering
(b) refraction and absorption
(c) dispersion and focusing
(d) total internal reflection and dispersion

252. As the sunlight passes through the atmosphere, the rays are scattered by tiny particles of dust, pollen, soot and other minute particulate matters present there. However, when we look up, the sky appears blue during mid-day because
(a) blue light is scattered most
(b) blue light is absorbed most
(c) blue light is reflected most
(d) ultraviolet and yellow component of sunlight combine

253. Consider the following statements
1. Clear sky appears blue due to poor scattering of blue wavelength of visible light.
2. Red part of light shows more scattering than blue light in the atmosphere.
3. In the absence of atmosphere, there would be no scattering of light and sky will look black.

Which of the statements given above is/are correct?
(a) Only 1 (b) 1 and 2
(c) Only 3 (d) All of these

254. We cannot see the details of a distant object because
(a) there is a defect in our eyes
(b) the light rays are absorbed and scattered in this way
(c) there is a limit of resolution for our eyes
(d) the objects are too far away

255. Match the following columns.

Column I	Column II
A. Myopia	1. Cylindrical
B. Hypermetropia	2. Bifocal
C. Presbyopia	3. Convex
D. Astigmatism	4. Concave

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	2	1	(b)	1	2	3	4
(c)	2	4	3	1	(d)	2	1	4	3

256. The principle of working of periscope is based on
(a) reflection only
(b) refraction only
(c) reflection and interference
(d) reflection and refraction

257. For a telescope, the focal lengths of two lenses are 30 cm and 10 cm. One acts as objective and other as eyepiece. The length of the telescope is
(a) 40 cm (b) 3 cm
(c) 200 cm (d) 30 cm

258. Consider the following statements
1. Periscope consists of two plane mirror inclined at an angle of 45°.
2. The principle of working of periscope is based upon reflection and refraction.

Which of the following statements is/are correct?
(a) Only 1 (b) 1 and 2
(c) Only 3 (d) None of these

Previous Years' Questions

259. Dispersion process forms spectrum due to white light falling on a prism. The light wave with shortest wavelength **2013 II**
(a) refracts the most
(b) does not change the path
(c) refracts the least
(d) is reflected by the side of the prism

260. A ray of white light strikes the surface of an object. If all the colours are reflected the surface would appear **2013 II**
(a) black (b) white (c) grey (d) opaque

261. Which one among the following colours has the highest wavelength? **2013 II**
(a) Violet (b) Green (c) Yellow (d) Red

262. No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be **2014 I**
(a) either plane or convex
(b) plane only
(c) concave (d) convex only

263. The position, relative size and nature of the image formed by a concave lens for an object placed at infinity are respectively **2014 I**
(a) at focus, diminished and virtual
(b) at focus, diminished and real
(c) between focus and optical centre, diminished and virtual
(d) between focus and optical centre, magnified and real

264. In the phenomenon of dispersion of light, the light wave of shortest wavelength is **2014 II**
(a) accelerated and refracted the most
(b) slowed down and refracted the most
(c) accelerated and refracted the least
(d) slowed down and refracted the least

265. The upper and lower portions in common type of bi-focal lenses are respectively **2014 II**
(a) concave and convex
(b) convex and concave
(c) both concave of different focal lengths
(d) both convex of different focal lengths

266. A person standing 1m in front of a plane mirror approaches the mirror by 40 cm. The new distance between the person and his image in the plane mirror is **2015 I**
(a) 60 cm (b) 1.2 m (c) 1.4 m (d) 2.0 m

267. The outside rear view mirror of modern automobiles is marked with warning 'objects in mirror are closer than they appear'. Such mirrors are **2015 II**
(a) plane mirrors
(b) concave mirrors with very large focal lengths
(c) concave mirrors with very small focal lengths
(d) convex mirrors

268. The focal length of the lens of a normal human eye is about **2015 II**
(a) 25 cm (b) 1 m
(c) 2.5 mm (d) 2.5 cm

269. A myopic person has a power of – 1.25 D. What is the focal length and nature of his lens? **2016 I**
(a) 50 cm and convex lens
(b) 80 cm and convex lens
(c) 50 cm and concave lens
(d) –80 cm and concave lens

Electric Current

270. Which of the following is a good conductor of heat but a bad conductor of electricity?
(a) Mica (b) Aluminium
(c) Mercury (d) Platinum

271. Consider the following statements
1. The direction of positive charges is same as direction of conventional current.
2. The rate of flow of electric current charges through any cross-section of conductor is called as electric current.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

272. The work done in moving a unit positive charge across two points in an electric circuit is a measure of
(a) potential difference
(b) electrostatic force
(c) current (d) power

273. Potential difference between two points of a wire carrying 2A current is 0.1 V. The resistance of the wire is
(a) 0.05 Ω (b) 0.5 Ω
(c) 5 Ω (d) 0.25 Ω

274. Match the following columns.

Column I	Column II
A. Electrical force between two charged particles	1. Volt
B. Electric charge	2. Newton
C. Electric potential	3. Farad
D. Electrical capacity	4. Coulomb

Codes

	A	B	C	D
(a)	2	4	1	3
(b)	2	4	3	1
(c)	1	2	3	4
(d)	2	1	3	4

275. In electric supply lines in India, which parameter is kept constant?
(a) Voltage (b) Current
(c) Frequency (d) Power

276. Which one of the following is not correctly matched?
(a) Voltmeter — Potential difference
(b) Potentiometer — EMF
(c) Ammeter — Electric current
(d) Meter bridge — Electrical resistance

277. A DC voltmeter is capable of measuring a maximum of 300 V. If it is used to measure the voltage across a device operating at 220 V AC supply, the reading of the voltmeter will be
(a) 0 V (b) 110 V (c) 220 V (d) 300 V

278. Galvanometer
1. is a device used to detect and measure electric current in a circuit.
2. can be converted into a voltmeter by connecting a very high resistance in its parallel.

Which of the following statements is/are correct?
(a) Only 1 (b) 1 and 2
(c) Only 2 (d) None of these

279. When you pull out the plug connected to an electrical appliance, you often observe a spark. To which property of the appliance is this related?
(a) Resistance (b) Inductance
(c) Capacitance (d) Wattage

280. Match the columns.

Column I		Column II	
A.	Voltage	1.	$\frac{q}{t}$
B.	Current	2.	$\rho\frac{l}{A}$
C.	Capacity	3.	$\frac{c}{v}$
D.	Resistance	4.	IR

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	1	3	2
(c)	3	4	2	1	(d)	4	3	2	1

281. Two resistances when connected in parallel give the resultant value of 2 Ω, when connected in series the value becomes 9 ohm. The value of each resistance is
(a) 1 Ω and 8 Ω (b) 2 Ω and 7 Ω
(c) 3 Ω and 3 Ω (d) 3 Ω and 6 Ω

282. Consider the following statements
1. The property of a substance by virtue of which it opposes the flow of current through it, is known as the resistance.
2. Resistance of a conductor is directly proportional to it's length.
3. Resistance of a conductor is directly proportional to area of cross-section.
4. Resistance of a substance is always equal to resistivity.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 4
(c) 3 and 4 (d) 1, 2 and 3

283. Consider the following statements
1. Electric appliances with metallic body e.g. heater, presses, etc., have three pin connections whereas, an electric bulb has a two pin connection.
2. Three pin connections reduce heating of connecting cables.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) 1 and 2 (d) None of these

284. An electric bulb is marked as 240 V, 60 W. The resistance of its filament is
(a) 940 Ω (b) 245 Ω
(c) 950 Ω (d) 960 Ω

285. A tube light works on the principle of
(a) chemical effect of current
(b) magnetic effect of current
(c) heating effect of current
(d) discharge of electrons through gases

286. The wire used in the filaments of household bulbs has
(a) high resistance, high melting point
(b) high resistance, low melting point
(c) low resistance, high melting point
(d) low resistance, low melting point

287. Safety fuses are integral part of electric installations and instruments. This is so because safety fuses
(a) block the passage of current due to increase in their resistance and saves it
(b) switch off the service of electric supply through relay action
(c) provide alternative path to excess current as does a shunt
(d) switch off the supply, if current beyond a certain limit flows through the circuit

288. In a filament type high bulb, most of the electric power consumed appear as
(a) visible light
(b) infrared rays
(c) ultraviolet rays
(d) fluorescent light

289. Why are inner lining of hot water geysers made up of copper?
(a) Copper has low heat capacity
(b) Copper has high electrical conductivity
(c) Copper does not react with steam
(d) Copper is good conductor of both heat and electricity

290. Consider the following statements
1. Fuse wire is made up of tin lead alloy.
2. Fuse wire have low resistivity.
3. Fuse wire have low melting point.
4. Fuse wire is used in parallel as a safety device in an electric circuit.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 2 and 4
(c) Only 4 (d) 1 and 2

291. A bird sitting on a high power line
(a) gets a fatal shock
(b) gets a mild shock
(c) gets killed instantly
(d) is not affected practically

292. The total resistance of 3 resistors, each of 3 Ω, connected in parallel, is
(a) 3 Ω (b) 2 Ω
(c) 1 Ω (d) 9 Ω

293. The circuit element where the impressed voltage is always in phase with the resulting current, is
(a) an ideal resistor
(b) an ideal coil
(c) an ideal capacitor
(d) an ideal transformer

294. The tape of a tape recorder is coated with
(a) zinc oxide
(b) mica
(c) copper sulphate
(d) ferromagnetic powder

295. Consider the following statements
1. Permanent magnets are usually made of alloys such as carbon steel, chromium steel, tungsten steel and alnico.
2. Alnico is alloy of gold, silver and zinc.